

SoK: Migrating Cryptocurrencies to the Post-Quantum Era – Challenges, Solutions, and Open Problems

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Abstract—The advent of quantum computing poses significant risks to the cryptographic foundations of cryptocurrency protocols. This work presents a systematization of knowledge (SoK) on the post-quantum readiness of these protocols by analyzing their underlying cryptographic primitives. We classify primitives into three categories: (i) post-quantum secure, (ii) not post-quantum secure but with practical post-quantum alternatives, and (iii) not post-quantum secure with no alternatives or only impractical ones. This taxonomy allows us to outline potential migration paths for these protocols, the challenges they are likely to face, and the open problems that remain for a successful post-quantum transition.

Index Terms—post-quantum, digital currencies,

I. INTRODUCTION

Cryptographic primitives are the mathematical and algorithmic foundations of modern digital security, providing essential functions such as confidentiality, integrity, authentication, and privacy. In the context of cryptocurrencies, these primitives underpin critical protocol components, enabling secure transactions, protecting user identities, and ensuring the correctness of distributed ledger operations. Commonly deployed primitives include digital signature algorithms (e.g., ECDSA, EdDSA, BLS), elliptic curve systems (e.g., secp256k1, Curve25519, Jubjub), zero-knowledge proofs, and key exchange protocols (e.g., Diffie–Hellman, ECDH). Each primitive offers unique security guarantees and performance trade-offs, yet exhibits varying levels of resistance against different types of attacks. The emergence of quantum computing introduces a paradigm shift in this landscape: algorithms such as Shor’s and Grover’s threaten to render many widely used public-key schemes insecure, particularly those relying on integer factorization and discrete logarithm problems. This growing threat has motivated the development of Post-Quantum Cryptography (PQC), which seeks to design cryptographic algorithms secure against both classical and quantum adversaries. For cryptocurrencies, this transition is especially critical, as their decentralized and immutable nature leaves no room for retroactive fixes once vulnerabilities are exploited. In this work, we analyze the cryptographic primitives employed by various cryptocurrency systems and classify them based on their resilience against quantum attacks. We further evaluate these systems in the context of known post-quantum alternatives, highlighting the gap between current deployments and quantum-secure architectures.

A. Requirements for a PQ-Secure Cryptocurrency Protocol

(A.1.) What are the Quantum Computers: Quantum computers utilize principles of quantum mechanics such as superposition and entanglement to process information in ways that classical computers cannot. Unlike classical bits, which can represent either 0 or 1, quantum bits (qubits) can represent multiple states simultaneously. This property enables quantum computers to solve certain complex problems much faster than classical computers.[1]

(A.2) Quantum Computer Algorithms

To ensure security against quantum computers, cryptocurrency systems need to withstand specific types of quantum attacks. Two such potential attack methods are described below

(A.2.1) Shor’s Algorithm: On a quantum computer, to factor an integer N , Shor’s algorithm runs in polynomial time, meaning the time taken is polynomial in $\log N$ [2]. It takes quantum gates of order

$$O((\log N)^2(\log \log N)(\log \log \log N))$$

using fast multiplication or even

$$O((\log N)^2(\log \log N))$$

utilizing the asymptotically fastest multiplication algorithm currently known due to Harvey and van der Hoeven [3], thus demonstrating that the integer factorization problem can be efficiently solved on a quantum computer and is consequently in the complexity class BQP.

This is significantly faster than the most efficient known classical factoring algorithm, the general number field sieve, which works in sub-exponential time:

$$O\left(e^{1.9(\log N)^{1/3}(\log \log N)^{2/3}}\right)$$

Shor’s Algorithm has suggested solutions to the factoring problem and the discrete logarithm problems. For this reason, the effectiveness of Shor’s algorithm and the problems it can solve are examined in our report.

(A.2.2) Grover’s Algorithm: Grover’s algorithm is a quantum algorithm that solves the unstructured search problem. If we are given a set consisting of N elements as a problem, and we want to find a single marked element within that set, Today’s computers must search through all N elements to find that element, which takes $O(N)$ time. Grover’s algorithm, however, can find that element in $O(\sqrt{N})$ time. Grover’s algorithm can

be used to break cryptographic keys and solve optimization problems. For this algorithm, it would be sufficient to double the key size and hash used. We did not mention key and hash problems in this report because security against quantum computers can be ensured this way.

II. REQUIREMENTS FOR A SECURE PQ PROTOCOL

In recent years, significant research has focused on quantum computers—machines that leverage quantum mechanical phenomena to solve mathematical problems that are infeasible for classical computers. The emergence of quantum computers poses a serious threat to many currently used public-key cryptosystems, which could potentially be broken by quantum attacks.

To address this challenge, the field of Post-Quantum Cryptography aims to develop cryptographic algorithms that remain secure against both classical and quantum adversaries. Recognizing the urgency, NIST has published standardized algorithms for various quantum-resistant cryptographic primitives. While many of these algorithms have already been released, ongoing efforts continue to refine and expand the suite of cryptographic primitives necessary for ensuring quantum-safe security in applications such as cryptocurrencies.

A. Abbreviations and Acronyms

Classical Signature: A mechanism that allows users to sign messages with their private keys and everyone to verify these signatures with their public keys. These signatures ensure that transactions on the network are secure, verifiable and immutable and has the following features

- Identity verification
- Data integrity
- Transaction authorization

BLS(Boneh–Lynn–Shacham) Signature: Enables multiple digital signatures to be combined under a single signature.

- Provides efficiency and scalability by reducing multiple signatures to a single short signature.
- Reduces data load on the network.

Ring Signature: A type of digital signature that allows a person from a group (ring) to sign a message, but hides which one signed it.

- Provides anonymity
- Preserves the validity of the signature while concealing the identity of the signer

SNARK (Succinct Non-interactive Argument of Knowledge): A structure that allows one party to prove that a calculation has been performed correctly with a very small and quickly verifiable proof. It is a non-interactive zero-knowledge proof.

- Proves the accuracy of transactions while protecting data privacy
- Provides scalable, anonymous, and fast transaction confirmation

Hash Puzzle : A computational process in Proof-of-Work (PoW) systems that searches for a hash meeting a specific

difficulty target, similar to solving a puzzle. It enables mining as a consensus mechanism and helps prevent spam and network attacks and has the following features:

- Provides a consensus mechanism (mining)
- Makes it difficult to spam and attack the network

Protocol Type: Refers to the category of a cryptocurrency protocol, which defines the fundamental operational principles of the system. Examples include:

Consensus Protocols – e.g., Proof of Work (PoW), Proof of Stake (PoS).

Layer 1 Protocols – Base blockchain infrastructures that handle transactions and security directly.

Layer 2 Protocols – Built on top of Layer 1 blockchains (e.g., Ethereum) to improve scalability and efficiency; examples include rollups and payment channels.

Zero-knowledge (ZK)-based protocols - Utilize zero-knowledge proofs to enhance privacy and security.

Understanding the type of protocol is essential to identify which technology addresses specific scalability, security, and privacy challenges in cryptocurrency systems.

Homomorphic Commitment: A structure that proves “commitment” to a value while concealing the content of the data. This commitment also supports operational features (homomorphic: addition/multiplication can be performed).

- It conceals the transaction amount and proves that it has been correctly added up.

III. EVALUATION OF QUANTUM SAFE CRYPTOGRAPHIC PRIMITIVES

Cryptocurrency protocols use many cryptographic primitives that are not quantum-safe, as explained above. For this reason, as this technology develops in the future, quantum computers will pose a major threat to the cryptocurrency market. The transition is essential and must be made as soon as possible. In **Table 1**, we analyzed the structure of cryptocurrencies that use 13 different popular cryptographic primitives and have different mechanisms. We classified them according to their structures and then classified them according to whether the cryptographic primitives they use are post-quantum safe or not. Our classification was as follows: if the primitive used had a post-quantum equivalent, we classified its transition as more feasible; if it had a post-quantum equivalent and was being used, we classified it accordingly; if it did not have a post-quantum equivalent, we indicated this in the table. Overall, looking at the table, the prevalence of these shortcomings indicates that cryptocurrencies are vulnerable to quantum computers.

A. Table Primitives

Our evaluation in the table was based on the use of classic signatures (ECDSA, EdDSA) in the protocol of the cryptocurrency under review. Some cryptocurrency protocols have used signatures such as Ring Signature and BLS instead of classic signatures for different purposes in their protocols. These signatures are shown separately in the table because they are used differently in protocols compared to classic signatures.

Protocol	Classical Signature	Aggregated Signature (BLS)	Ring Signature	SNARK	Hash Puzzle (in PoW)	Protocol Dependencies & Assumptions	Protocol Type	Homomorphic Commitment
Bitcoin	!				✗		Layer 1	
Ethereum	!	✗					Layer 1	
Zcash	!			!	✗	Bulletproof	Layer 1	!
Zether	!				✗	Σ -Bullets, Ethereum Consensus	Layer 1	!*
Monero			!		✗	Bulletproofs	Layer 1	!
Aleo	!			!			Layer 1	
Arbitrum	!					Ethereum Consensus	Optimistic Layer 2	
Optimism	!					Ethereum Consensus	Optimistic Layer 2	
StarkNet	!			✓		Ethereum Consensus	Zk layer 2	
Scroll	!			!		Ethereum Consensus	Zk layer 2	
ZKsync	!			!		Ethereum Consensus	Zk layer 2	
Polygon	!			!		Ethereum Consensus	Zk layer 2	
Aztec	!			!		Ethereum Consensus	Zk layer 2	

Table 1

✓	Post Quantum Secure
!	Not PQ secure, but there is an equivalent to PQ secure
✗	There is no equivalent to the efficient PQ secure
!*	Homomorphic encryption

Many cryptocurrencies have integrated zero-knowledge proof systems into their protocols. SNARKs are used for different purposes within protocols.

B. Explanation of Cryptocurrencies According to the Post-Quantum Secure Table

Bitcoin: These cryptocurrency transactions rely on classical digital signatures, which are vulnerable to Shor's algorithm and therefore lack quantum security. To address this issue, the signatures can be replaced with quantum-secure digital signature schemes standardized by NIST (e.g., Falcon, Dilithium). Bitcoin employs a Proof-of-Work (PoW) consensus mechanism, which is inherently resistant to quantum attacks. However, in the current literature, there is no fair and practical network system that utilizes this mechanism without benefiting from the computational advantages of quantum computing.

Ethereum: This cryptocurrency is a layer 1 protocol, similar to Bitcoin. Layer 1 protocols are used to describe cryptocurrencies that perform all transactions on their own blockchain. Ethereum uses a classic signing algorithm to sign

transactions, which can be replaced with quantum-safe signing algorithms. The biggest obstacle to Ethereum transitioning to a quantum-safe protocol is the use of the BLS signature. The BLS signature enables the aggregation of Ethereum validators' signatures and is a highly effective and efficient signing algorithm for Ethereum network. However, there is no quantum-safe alternative as efficient as the BLS signature, and efforts to find an equally efficient alternative signature scheme are ongoing [4].

Zether: This cryptocurrency uses ElGamal homomorphic encryption to hide the amount of users. As shown in the table, El-Gamal performs homomorphic addition, but it does so on encrypted data. This encryption is not quantum-safe, and there is currently no equally efficient quantum-safe alternative in the literature. Unlike Bulletproof [5], Zether uses Σ -Bullet. It is used to verify that ElGamal encrypted data falls within a certain range.

Monero: This cryptocurrency protocol employs ring signatures instead of classical digital signature algorithms. Post-quantum research has been conducted on Efficient Linkable Ring Signatures as a quantum-resistant counterpart to this signature scheme [6]. Monero utilizes the Pedersen commitment scheme to conceal user transaction amounts. This scheme is homomorphic with respect to aggregation, allowing encrypted amounts to be combined. To prove that these amounts lie within a specific range, Bulletproofs are employed. Although homomorphic commitment schemes like Pedersen are not quantum-safe, lattice-based homomorphic commitment schemes have been proposed in the literature as potential post-quantum alternatives [7].

Aleo: Aleo is classified as a Layer 1 cryptocurrency

that leverages zero-knowledge proof protocols to ensure transaction privacy. When a user initiates a transaction, they generate a zero-knowledge proof attesting to the validity of their transaction without revealing sensitive details. This proof is then submitted to the network validators (verifiers). Validators first verify the zero-knowledge proof and subsequently authenticate the transaction using a classical digital signature. The classical digital signature currently in use can be replaced with NIST-standardized quantum-safe signature schemes to enhance post-quantum security. Additionally, the zero-knowledge proofs generated by the prover, typically SNARKs, have quantum-safe counterparts known as STARKs, which can be adopted to further strengthen the protocol's resistance against quantum attacks.

Arbitrum: Arbitrum is an Ethereum Layer 2 scaling solution designed to reduce the computational load on the Ethereum mainnet. It achieves this by processing transactions off-chain and submitting succinct proofs of transaction validity to Ethereum validators on Layer 1. Transactions within Arbitrum are currently authorized using classical digital signatures, which can be upgraded to quantum-safe signature schemes to enhance security against quantum attacks. The validity proofs employed in Arbitrum rely on SNARKs (Succinct Non-Interactive Arguments of Knowledge), which are not inherently quantum-secure. However, these can be replaced with STARKs (Scalable Transparent Arguments of Knowledge), offering quantum-resistant proof systems that maintain the integrity and efficiency of transaction verification while improving post-quantum security.

Optimism: This cryptocurrency is also an optimistic Layer 2 solution and employs the same cryptographic primitives as Arbitrum. Therefore, the quantum-safe solutions applicable to Arbitrum are equally valid for this cryptocurrency.

C. ZK-Layer 2 Cryptocurrency Quantum Safety

ZK-Layer 2 cryptocurrencies is an Ethereum Layer 2 solution that relies on a mechanism where proofs of transactions are verified by Ethereum validators on the Layer 2 network to reduce the computational load on the Ethereum blockchain. Transactions are currently signed using classical digital signatures, which are vulnerable to quantum attacks but can be replaced with quantum-safe signature algorithms compliant with NIST standards. Transaction validity proofs are generated using SNARKs, which are not quantum-safe; however, these can be replaced by their quantum-resistant counterparts, STARKs, to improve post-quantum security. In this section, zk-Layer 2 solutions are evaluated collectively in terms of quantum security due to their use of similar cryptographic primitives with only subtle differences. Among these, StarkNet, Scroll, zkSync, Polygon, and Aztec stand out as prominent Layer 2 solutions for Ethereum. Scroll and zkSync, in particular, share closely related designs and cryptographic frameworks. While most of these solutions currently rely on classical signature schemes that are vulnerable to quantum attacks, they have the potential to adopt quantum-

safe signature schemes standardized by NIST. These Layer 2 solutions submit zero-knowledge proofs to the Ethereum Layer 1 blockchain for transaction validation, where the proofs are verified by Ethereum validators. Except for StarkNet, which employs post-quantum secure STARK proofs, the others—Scroll, Polygon, and Aztec—utilize SNARKs that are not quantum-safe but could be replaced with STARKs for enhanced security. It is important to note that all these zk-Layer 2 cryptocurrencies inherit Ethereum's consensus mechanism, which uses BLS signatures that are themselves not quantum-safe, highlighting a critical challenge in achieving full post-quantum security for the entire system. Moreover, zk-Layer 2 solutions implement additional SNARK and proof verification checks directly within their smart contracts to ensure transaction validity and privacy guarantees, adding an extra layer of security and trustworthiness to the system.

Aztec: Aztec is recognized as the first Layer 2 solution on Ethereum to provide privacy-preserving features at scale. It enables confidential transactions that hide transaction amounts and participant identities by using zero-knowledge proofs. Its design prioritizes both scalability and privacy, making it a pioneering protocol in the zk-Layer 2 ecosystem. Unlike other Layer 2 solutions, it does not possess cryptographic primitives that are secure against quantum computers.

IV. CONCLUSION

Blockchain technology fundamentally relies on cryptographic primitives such as hash functions, zero-knowledge proofs, and digital signatures. Most current blockchain platforms employ elliptic curve cryptography to generate digital signatures. However, the emergence of quantum computers and algorithms like Shor's algorithm threatens these schemes by efficiently solving their underlying hard problems[8]. SNARK proofs widely used in cryptocurrencies are also not quantum-safe, as they rely on discrete logarithms and elliptic curves. Post-quantum alternatives, such as STARKs, offer promising solutions to enhance quantum resistance. In this study, we evaluated various cryptocurrencies with Layer 1 and Layer 2 solutions in terms of their quantum resilience. Our analysis reveals that most cryptocurrencies currently lack sufficiently quantum-secure primitives[9]. Conversely, over %50 of the TLS internet protocol ecosystem has transitioned to post-quantum key encapsulation mechanisms (KEMs). This is primarily because quantum-safe equivalents exist for both public key encryption and digital signature algorithms, facilitating a smoother transition. Cryptocurrencies, however, face additional challenges. Layer 2 solutions are dependent on Ethereum Layer 1's BLS signatures, and the Proof-of-Work consensus mechanism is hash-based. Due to these dependencies and the cryptographic primitives involved, transitioning cryptocurrencies to post-quantum secure algorithms is complex and challenging. Therefore, achieving post-quantum security in cryptocurrencies will be a longer and more difficult process compared to internet protocols.

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